

Udit Mittal

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Summary

Full-Stack Software Engineer — Final-year Computer Science undergraduate at VIT (CGPA 8.51/10, VITEEE AIR 162) with two software engineering internships. Ships production web applications end to end: REST API design, relational and NoSQL data modelling, and cloud-native deployment.

Technical Skills

Languages: C++, Java, Python, C, SQL, TypeScript, JavaScript
Backend: Spring Boot, FastAPI, REST APIs, Microservices, AWS Lambda, API Gateway
Frontend: React, Next.js, HTML, CSS, Tailwind CSS
Databases: MySQL, PostgreSQL, NoSQL (Amazon DynamoDB), Query Optimization
Cloud & DevOps: AWS (Lambda, S3, DynamoDB, API Gateway), Docker, GitHub Actions
CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Computer Networks
Tools & Practices: GitHub, Postman, JIRA, Agile/Scrum, Design Patterns, System Design

Projects

Real-Time Energy Management and Autonomous Load Optimization (in progress) Jun 2026 – Present
Python, FastAPI, Machine Learning, React, Next.js, TypeScript, IoT, MQTT, AWS

- Designing an AI-driven industrial energy platform for real-time monitoring and autonomous load optimization at 10,000+ IoT sensor readings per minute.
- Developing 14 FastAPI REST endpoints for IoT ingestion, analytics, and JWT authentication, sustaining 500+ concurrent requests in load testing.
- Training LSTM and Isolation Forest models (scikit-learn, TensorFlow) for demand forecasting and anomaly detection, reaching 92% validation accuracy.
- Building a real-time **data engineering pipeline** (MQTT streaming ingestion) at sub-second end-to-end latency, feeding a React/Next.js dashboard of live energy KPIs.

FileFinder — Serverless File Management System Jan 2025 – Apr 2025
React, Next.js 14, TypeScript, Tailwind CSS, AWS Lambda, Amazon S3, DynamoDB, API Gateway

- Architected a serverless platform on AWS Lambda, S3, DynamoDB, and API Gateway with a Next.js 14 frontend, scaling 0 to 1,000+ concurrent requests at near-zero idle cost.
- Implemented 4 REST APIs for upload, search, download, and deletion at sub-300ms response times via bundle-size optimization.
- Integrated S3 pre-signed URLs for direct client-side uploads, bypassing the API layer and cutting upload latency 50%.
- Modeled a DynamoDB single-table schema with global secondary indexes, enabling single-digit-ms search over 10,000+ records.

Experience

Software Development Intern May 2026 – Jun 2026
Core Integra Global Services Pvt. Ltd. India

- Delivered 8+ full-stack features for the Ctrl+F web application (Java, Spring Boot, React), resolving 25+ defects and cutting reported production issues 30%.
- Designed and integrated REST API contracts between Spring Boot services and the React frontend with a 6-member Agile team, cutting integration rework 40%.
- Managed 40+ pull requests on a feature-branch workflow, reviewing peer code and reducing merge conflicts 35%.

Summer Intern May 2025 – Jun 2025
Core Integra Global Services Pvt. Ltd. India

- Engineered 12 RESTful APIs in Java and Spring Boot serving 5,000+ daily requests at sub-200ms latency.
- Optimized 20+ SQL queries through indexing, join restructuring, and execution-plan analysis, reducing average query latency from 1.2s to 300ms.
- Shipped 4 Spring Boot backend modules and resolved 30+ defects, cutting the open bug backlog 45% ahead of release.

Education

Vellore Institute of Technology Aug 2023 – May 2027
B.Tech in Computer Science and Engineering — CGPA: 8.51/10 Vellore, India

Shemford Futuristic School Mar 2022 – May 2023
Senior Secondary School (CBSE) Saharanpur, India

Leadership & Achievements

- All India Rank 162**, VIT Engineering Entrance Examination (VITEEE).
- Head Boy**, Shemford Futuristic School (2022–23) — led the student council representing 500+ students.
- Event Coordinator**, Guest Care Committee, Gravitas 2025, VIT — ran guest logistics across a 3-day national technical festival.
- Bharat Scouts & Guides** (3 years) — selected to represent the school at a national engagement with the President of India (Nov 2022).
- Certification: C++ Programming — Beginner to Advanced (Deep Dive in C++).